

Education

Rensselaer Polytechnic Institute, Troy, NY

Masters of Science in Mechanical Engineering, GPA 3.92

Bachelors of Science in Mechanical Engineering, Minor in Economics, GPA 3.53, Cum Laude

Spring 2026-Fall 2026

Fall 2022-Fall 2025

Professional Interests

Interested in research-oriented engineering roles focused on advancing computational modeling and simulation, high-performance computing, robotics, controls, and machine learning.

Experience

Graduate Research, Center for Scientific Computation at Rensselaer Polytechnic Institute, Troy, NY	Summer 2025-Present
Developing software to convert complex nuclear geometries between Constructive Solid Geometry (CSG) and Boundary Representation (BREP) modeling regimes, including ITER tokamak and light-water reactor models.	Developing GPU-accelerated neutronics workflows coupling OpenMC with unstructured meshes for neutron flux and volumetric heating simulations.
Product Security Intern, GE Vernova - Wind, Schenectady, NY	Summer 2025
Assessed industrial network vulnerabilities, proposing network segmentation and micropatching solutions for critical systems.	Developed role-based cybersecurity training and evaluated secure-coding tools to strengthen internal development practices.
Control Systems Intern, GE Vernova - Gas Power, Schenectady, NY	Summer 2024
Developed, tested, and qualified industrial control and digital networking devices for gas turbine customer environments.	Simulated customer network configurations, integrated controls hardware/software

Projects

Cable-in-Conduit Thermal Analysis - Introduction to Computational Fluid Dynamics, Rensselaer Polytechnic Institute	Spring 2026
- Iteratively developed increasingly complex CAD models of representative Cable-in-Conduit (CIC) sections used to cool superconducting magnetic strands in tokamak fusion reactors. - Applied FEM/FEA unstructured meshes and thermal-fluid boundary conditions, then conducted high-fidelity CFD simulations of conjugate heat transfer and fluid dynamics within CIC systems. - Evaluated the ability of supercritical helium cooling to maintain superconducting magnetic strands within operating conditions under thermal loads.	
Robotic Path Planning with Neural Networks - Scientific Machine Learning Final Project, Rensselaer Polytechnic Institute	Spring 2026
- Developed and evaluated CNN-based heuristics for robotic navigation, testing whether neural networks could improve upon Euclidean distance while maintaining admissibility across varied 2D obstacle environments. - Trained and compared CNNs using MSE and asymmetric MAE loss functions, and presented findings to peers and faculty.	
HPC Parallel Image Recreation and Rendering - Parallel Computing Final Project, Rensselaer Polytechnic Institute	Spring 2026
- Collaborated with a team of graduate students to develop a massively parallel, GPU-accelerated genetic algorithm for image reconstruction. - Deployed the algorithm on RPI's AiMOS supercomputer and evaluated strong/weak scaling across multi-GPU and multi-node MPI configurations. - Analyzed performance results and co-authored a conference-style paper documenting the methodology, scalability, and findings.	
UAV Wing Spar Optimization Under Uncertainty - Numerical Design Optimization Project, Rensselaer Polytechnic Institute	Fall 2025
- Optimized a carbon-fiber UAV wing spar under uncertain aerodynamic loading using MATLAB fmincon, surrogate modeling, and Latin hypercube sampling. - Reduced spar mass by 70.68% while satisfying six-sigma probabilistic stress constraints under complex aerodynamic loading.	

Leadership and Activities

Pi Lambda Phi Fraternity - President	Fall 2024 - Fall 2025
- Led a seven-member executive board and 35-45-member chapter, overseeing operations across finance, housing, recruitment, risk management, alumni relations, fundraising, and member development. - Standardized leadership transitions and position expectations through documented procedures, mandatory transition meetings, and performance incentives. Oversaw ~\$20K in facility upgrades and improvements.	
Rensselaer Mens Rugby - Captain, Head Coach	Spring 2025 - Fall 2025, Fall 2027
- Led three weekly practices, team meetings, and film sessions while establishing team goals, expectations, and development plans. - Mentored players into future leadership roles, leading the team to its first National Collegiate Rugby 7s appearance and a Liberty League 15s runner-up finish.	
Rensselaer Mens Club Basketball - Vice President, Treasurer	Spring 2025 - Fall 2025
- Developed an executive leadership framework, standardized board transitions, and established documentation systems to improve organizational continuity. - Coordinated practices, game logistics, and player development while helping lead the team to its first undefeated season and second regional playoff appearance.	

Skills

Languages & Tools: Python, MATLAB, C++, C, CUDA, MPI, PyTorch, ROS2, Linux, Simulink, Siemens NX, Altair CFD, Slurm, FreeCAD, Gmsh

Engineering: CFD, FEM/FEA, HPC, GPU Computing, Scientific ML, Numerical Optimization, Surrogate Modeling, Dynamic Systems, Control Systems, LQR, Particle Transport, Unstructured Meshing